CASE 3
ACTIVATION-INDUCED CYTIDINE DEAMINASE DEFICIENCY

An intrinsic B-cell defect prevents immunoglobulin class switching.

Immunoglobulin class switching, also known as isotype switching, is a complex process. As we saw in Case 2, after antigen has been encountered and the mature B cell is activated, the rearranged immunoglobulin heavy-chain variable (V) region DNA sequence can become progressively associated with different constant (C)-region genes by a form of somatic recombination (see Fig. 2.2). This process of class-switch recombination requires the engagement of a specific biochemical machinery in activated B lymphocytes, which works most efficiently after the B cell has received signals from activated helper T cells, such as those delivered by the interaction of CD40 ligand (CD40L) on the T cell with CD40 on the B cell (see Fig. 2.3). In Case 2, we saw that a failure of class switching, leading to a hyper IgM syndrome characterized by production of IgM and IgD but not the other isotypes, can result from a defect in the CD40L gene and a consequent absence of functional CD40L protein. CD40L is encoded on the X chromosome and so this type of hyper IgM syndrome is seen only in males. Hyper IgM syndrome is, however, also encountered in females, in whom it has an inheritance pattern in many families that suggests an autosomal recessive inheritance (Fig. 3.1). As we saw in Case 2, hyper IgM due to CD40 deficiency has an autosomal recessive inheritance, and yet another cause of autosomal recessive hyper IgM syndrome has been discovered

TOPICS BEARING ON THIS CASE:

Antibody isotypes and classes

Isotype or class switching

Somatic hypermutation

This case was prepared by Raif Geha, MD, in collaboration with Ari Fried, MD.

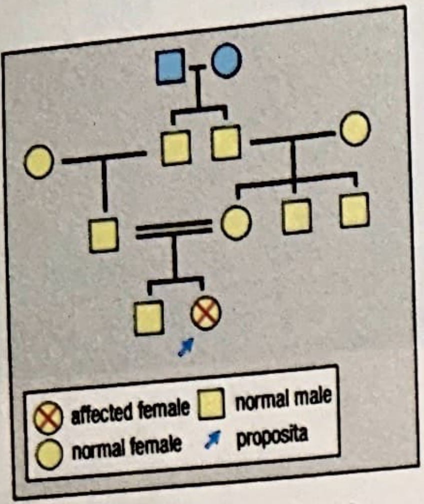

Fig. 3.1 A pedigree of a family with AID deficiency. Because the parents of the affected child (the proposita) show no signs of disease themselves, the defective AID gene must be recessive. The parents both carry a single copy of the defective gene, and, because they are first cousins, the most likely source is their shared grandfather or grandmother (blue symbols). If one of these were heterozygous for AID deficiency, they could have transmitted the defective gene to both their sons. The fact that the affected child is a girl, and that neither her father nor any other males in her extended family show signs of disease, indicates that the gene is carried on an autosome and not the X chromosome.

more recently—a defect in the B cell's biochemical pathways responsible for the class-switch recombination process itself.

The biochemical events underlying class switching have been quite well clarified. While studying a cultured B-cell line that was being induced to undergo class switching from IgM to IgA synthesis, immunologists in Japan observed a marked upregulation of the enzyme *a*ctivation-*i*nduced cytidine *d*eaminase (AID). This enzyme converts cytidine to uridine, and it is now known to trigger a DNA breakage and repair mechanism that underlies class-switch recombination. The contribution of AID to class switching was confirmed by 'knocking out' the gene encoding AID by homologous recombination in mice; the mutant animals developed hyper IgM syndrome and were unable to make IgG, IgA, or IgE.

The defective gene in some autosomal recessive cases of hyper IgM syndrome was mapped in several informative families to the short arm of chromosome 12, to a region corresponding with the region containing the *AID* gene in mice. This prompted the search for a link between hyper IgM syndrome and AID deficiency in humans, and several cases of the autosomal recessive form were found to have mutations in the *AID* gene.

The case of Daisy Miller: a failure of a critical B-cell enzyme.

Five-year-old girl with repeated ear infections.

At 3 years old, Daisy Miller was admitted to the Boston Children's Hospital with pneumonia. Her mother had taken her to Dr James, a pediatrician, because she had a fever and was breathing fast. Her temperature was high, at 40.1°C, her respiratory rate was 40 per minute (normal 20), and her blood oxygen saturation was 88% (normal >98%). Dr James also noticed that lymph nodes in Daisy's neck and armpits (axillae) were enlarged. A chest X-ray was ordered. It revealed diffuse consolidation (whitened areas of lung due to inflammation, indicating pneumonia) of the lower lobe of her left lung, and she was admitted to the hospital.

Daisy had had pneumonia once before, at 25 months of age, as well as 10 episodes of middle-ear infection (otitis media) that had required antibiotic therapy. Tubes (grommets) had been placed in her ears to provide adequate drainage and ventilation of the ear infections.

In the hospital a blood sample was taken and was found to contain 13,500 white blood cells ml^{-1}, of which 81% were neutrophils and 14% lymphocytes. A blood culture grew the bacterium *Streptococcus pneumoniae*.

High IgM; IgG very low; no IgA. Hyper IgM syndrome?

Because of Daisy's repeated infections, Dr James consulted an immunologist. She tested Daisy's immunoglobulin levels and found that her serum contained 470 mg dl^{-1} of IgM (normal 40–240 mg dl^{-1}), undetectable IgA (normal 70–312 mg dl^{-1}), and 40 mg dl^{-1} of IgG (normal 639–1344 mg dl^{-1}). Although Daisy had been vaccinated against tetanus and *Haemophilus influenzae*, she had no specific IgG antibodies against tetanus toxoid or to the polyribosyl phosphate (PRP) polysaccharide antigen of *H. influenzae*. Because her blood type was A, she was tested for anti-B antibodies (isohemagglutinins). Her IgM titer of anti-B antibodies was positive at 1:320 (upper limit of normal), whereas her IgG titer was undetectable.

Defective CD40 or AID? Order DNA test.

Daisy was started on intravenous antibiotics. She improved rapidly and was sent home on a course of oral antibiotics. Intravenous immunoglobulin (IVIG) therapy was started, which resulted in a marked decrease in the frequency of infections.

Analysis of Daisy's peripheral blood lymphocytes revealed normal expression of CD40L on T cells activated by anti-CD3 antibodies, and normal expression of CD40 on B cells (Fig. 3.2). Nevertheless, her blood cells completely failed to secrete IgG

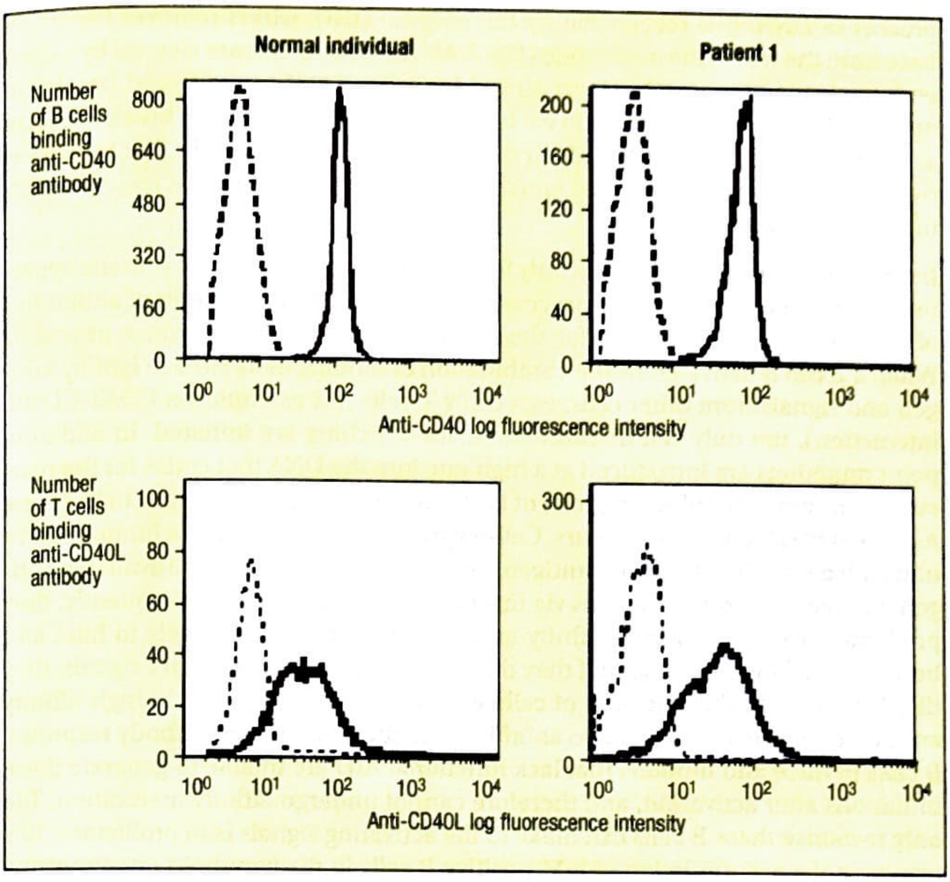

Fig. 3.2 Flow cytometric analysis showing normal expression of CD40 and CD40L in a patient with AID deficiency. Top row: CD40 measured by the binding of fluorescently tagged anti-CD40 antibodies to CD40 on B cells from (left panel) a normal individual and (right panel) a patient with AID deficiency. Bottom row: measurement of CD40L on T cells from (left panel) a normal individual and (right panel) the same patient activated *in vitro* with the mitogen phorbol ester (PMA) and ionomycin.

and IgE after stimulation with anti-CD40 antibody (to mimic the effects of engagement of CD40L) and interleukin-4 (IL-4), a cytokine that also helps to stimulate class switching, although the blood cells proliferated normally in response to these stimuli (Fig. 3.3). cDNAs for CD40 and for the enzyme activation-induced cytidine deaminase (AID) were made and amplified by the reverse transcription–polymerase chain reaction (RT–PCR) on mRNA isolated from blood lymphocytes activated by anti-CD40 and IL-4. Sequencing of the cDNAs revealed a point mutation in the *AID* gene that introduced a stop codon into exon 5, leading to the formation of truncated and defective protein. The CD40 sequence was normal.

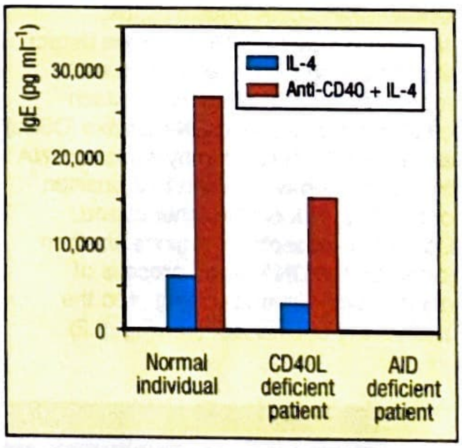

Fig. 3.3 Comparison of class switching, as judged by IgE secretion by B cells from a normal individual, a patient with CD40L deficiency, and a patient with AID deficiency. IgE production from peripheral blood mononuclear cells *in vitro* was measured after stimulation with IL-4 alone or stimulation with anti-CD40 and IL-4. The negative control was stimulation by medium alone (not shown), which produced barely detectable levels of IgE. The anti-CD40 and IL-4 together can compensate to some degree for the lack of stimulation of B cells by CD40L on T cells, but cannot compensate at all for the defect in AID.

Activation-induced cytidine deaminase deficiency (AID deficiency).

It is now apparent that there are several distinct phenotypes of hyper IgM syndrome, resulting from different genetic defects. One phenotype, which is caused by defects in the genes encoding CD40 or CD40L (see Case 2), manifests itself as susceptibility to both pyogenic and opportunistic infections. Another phenotype, which results from defects in the *AID* gene or in the gene encoding uracil-DNA glycosylase (UNG), another DNA repair enzyme involved in class-switch recombination, resembles X-linked agammaglobulinemia in that these patients have an increased susceptibility to pyogenic infections only.

When CD40 and the IL-4 receptor on B cells are ligated by CD40L and IL-4, respectively, the *AID* gene is transcribed and translated to produce AID protein. At the same time, transcription of cytidine-rich regions at specific isotype-switch sites is induced, which involves separation of the two DNA strands at these sites. AID, which can deaminate cytidine in single-stranded DNA only, then proceeds to convert the cytidine at the switch sites to uridine. Because uridine is not normally

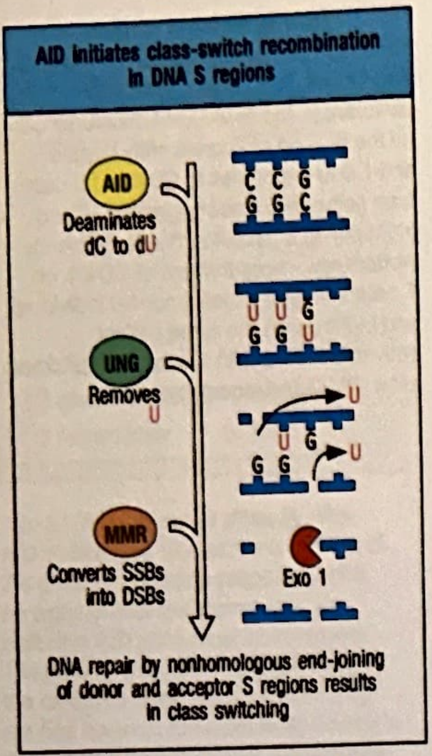

Fig. 3.4 AID initiates class-switch recombination. In the S regions upstream of the μ C gene and one of the other heavy-chain C genes, the enzyme activation-induced cytidine deaminase (AID) deaminates the cytosine in deoxycytidines (dC), resulting in uracil bases. Most of the uracil is then excised by the enzyme uracil-DNA glycosylase (UNG). The abasic sites are cleaved by endonuclease (not shown), creating single-strand DNA breaks (SSBs). Mismatch repair (MMR) enzymes detect the remaining U–G mismatches and recruit the exonuclease Exo1, which creates double-strand DNA breaks (DSBs) at each of the S regions by excising DNA from a nick on one strand to a position opposite a nick on the other strand. Donor and acceptor S regions are then joined by the DNA repair process of nonhomologous end-joining, and the intervening DNA is lost (see Fig. 2.2).

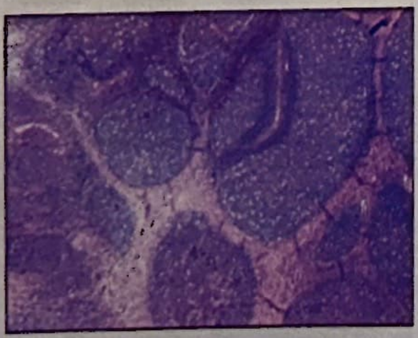

Fig. 3.5 Histology of a lymph node from a patient with AID deficiency stained with hematoxylin and eosin. Note the large follicles.

present in DNA, it is recognized by the enzyme UNG, which removes the uracil base from the rest of the nucleotide (Fig. 3.4). The abasic sites are cleaved by a DNA endonuclease, resulting in single-strand DNA breaks. If single-strand breaks on opposite DNA strands are near to each other, a double-strand DNA break will form at each switch region. It is the repair of these double-strand breaks that brings the two switch regions together and joins a new C-region gene to the V region, resulting in class switching.

Interestingly, AID is required not only for class switching but also for somatic hypermutation in B cells. This is the process that underlies the production of antibodies of increasingly higher affinity for the antigen as an immune response proceeds. When a B cell is activated by the combination of binding of its surface IgM by antigen and signals from other cells, especially T cells (for example via CD40–CD40L interaction), not only cell division and class switching are initiated. In addition, point mutations are introduced at a high rate into the DNA that codes for the rearranged immunoglobulin V regions of both the heavy-chain and light-chain genes. A process of selection then occurs. Cells expressing mutated surface immunoglobulin with a stronger affinity for antigen compete best for binding the available antigen and receive stronger signals via this new antigen receptor; consequently, they proliferate. Cells with lower-affinity immunoglobulins are less likely to bind and be stimulated by antigen, and if they do not receive these stimulatory signals, they die. This leads to the selection of cells with mutations that result in high-affinity antibodies, a process referred to as 'affinity maturation' of the antibody response. B cells in mice and humans that lack functional AID are unable to generate these mutations after activation, and therefore cannot undergo affinity maturation. The only response these B cells can make to the activating signals is to proliferate. This results in the accumulation of IgM-positive B cells in the lymphoid organs, giving rise to an enlarged spleen (splenomegaly) and enlarged lymph nodes (lymphadenopathy) (Fig. 3.5).

Another DNA repair mechanism, called mismatch repair (MMR), has a primary role in the detection and repair of DNA mismatches during meiosis. MMR has recently been found also to contribute to class-switch recombination downstream of AID and UNG. Proteins involved in MMR, namely Msh2, Msh6, Mlh2, and Pms2 build a complex that recognizes the U–G mismatches created by AID and, with the help of the exonuclease Exo1, excises the DNA from the nearest single-strand break past the site of the mismatch, creating a double-strand break. Thus, the role of MMR in class switching is thought to be the formation of double-strand breaks if the single-strand breaks introduced by AID and UNG on opposite DNA strands are not near enough to form one. Mice deficient in any of the MMR proteins or in Exo1 have a twofold to fivefold reduction in isotype switching. Deficiency of the MMR components Pms2 and Msh6 have been found to cause a distinct phenotype of class-switch recombination deficiency in humans. In addition to recurrent infections, patients with Pms2 deficiency have a high rate of cancer, as a result of generally defective DNA repair.

Questions.

1 *Daisy and other patients with hyper IgM syndrome caused by a mutation in AID do not seem to suffer from opportunistic infections, such as Pneumocystis jirovecii and Cryptosporidium, which are characteristic of CD40L deficiency. Why?*

2 *A 3-year-old girl presents with hyper IgM syndrome, recurrent bacterial infection, and a history of Pneumocystis jirovecii pneumonia. Her CD40L and AID genes are normal. What is the potential gene defect in this patient?*

3 A 6-year-old girl presents with hyper IgM syndrome, recurrent bacterial infection, and no history of opportunistic infections. Her CD40 and AID gene sequences are normal. What is the potential gene defect in this patient?

4 Can you think of a simple test to distinguish hyper IgM syndrome caused by an intrinsic B-cell defect from that caused by CD40L deficiency without recourse to DNA sequencing?

5 Dr James noted that Daisy had enlarged lymph nodes. Patients with CD40L deficiency do not have enlarged lymph nodes. Can you explain this difference?